

Miguel Esparza

Ph.D. in Electrical Engineering

Guadalupe, N.L., México

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Professional Summary

Senior Firmware Engineer with extensive experience in ARM Cortex-M architectures (M0/M3/M4) and Real-Time Operating Systems. Expert in designing CI/CD pipelines for embedded targets, developing low-level drivers, and implementing control algorithms. Skilled in C/C++, Python, and Docker for HIL testing environments.

Professional Experience

R&D Firmware Engineer @ Dematic

Feb 2023 - Present

Lead firmware architect for proprietary industrial devices, focusing on modular design and testability.

- Designed and implemented CI/CD pipelines using Docker containers for Hardware-in-the-Loop (HIL) testing.
- Developed optimized low-level drivers for Profinet, CAN, and UART communications.
- Implemented secure bootloader mechanisms and firmware update strategies for ARM microcontrollers.

R&D Manager (Technical Lead) @ Technologies Delee

Aug 2017 - Feb 2023

Led the firmware architecture design for a suite of biomedical diagnostic devices.

- Architected the firmware for six medical devices using STM32 microcontrollers.
- Developed precise flow control algorithms and power management systems.
- Mentored the engineering team in best coding practices and version control workflows.

Head Applications Engineer @ Solintec

Dec 2011 - Aug 2017

Developed automation logic and control software for industrial machinery.

- Programmed PLCs and HMI interfaces for over 20 automation projects.
- Integrated vision systems algorithms for quality control applications.

Technical Skills

Core Firmware:

C/C++, RTOS, Data Structures, OOP, Design Patterns

DevOps & Tools:

Git, Docker, Jenkins, Python scripting, Unit Testing

Embedded Systems:

ARM Cortex-M, Bootloaders, Peripherals (CAN, UART, SPI), JTAG

Control:

PID Control, Digital Signal Processing, Matlab/Simulink

Education

Ph.D. in Electrical Engineering

2013 - 2016

Universidad Autónoma de San Luis Potosí – Thesis: Modeling of VSC based systems. Research Internship at Aalborg University, Denmark.

M.E. in Electrical Engineering

2010 - 2012

UASLP – Thesis: FPGA-Based System for Induction Motors.

B.Eng. Electronic Engineering

2001 - 2006

Instituto Tecnológico de Durango – Ranked top 10 in class.

Selected Publications

- [1] C. Gurrola-Corral, J. Segundo, M. Esparza, and R. Cruz, “Optimal LCL-filter design method for grid-connected renewable energy sources,” *International Journal of Electrical Power and Energy Systems*, vol. 120, p. 105998, 2020, doi: 10.1016/j.ijepes.2020.105998.
- [2] J. Yee *et al.*, “Characterization of a novel automated microfiltration device for the efficient isolation and analysis of circulating tumor cells from clinical blood samples,” *Scientific Reports*, vol. 10, p. 7543, 2020, doi: 10.1038/s41598-020-63672-7.
- [3] J. Segundo, A. Bayo-Salas, M. Esparza, J. Beerten, and P. Gómez, “Frequency Domain Methods for Accuracy Assessment of Wideband Models in Electromagnetic Transient Stability Studies,” *IEEE Transactions on Power Delivery*, vol. 35, pp. 71–83, 2020, doi: 10.1109/TPWRD.2019.2927171.
- [4] C. Aguilar-Avelar *et al.*, “Fully Embedded Flow Control Device for Microfluidic Applications,” *IEEE Latin America Transactions*, vol. 18, pp. 446–454, 2020, doi: 10.1109/TLA.2020.9085302.
- [5] C. Aguilar-Avelar *et al.*, “High-Throughput Automated Microscopy of Circulating Tumor Cells,” *Scientific Reports*, vol. 9, pp. 1–9, 2019, doi: 10.1038/s41598-019-50241-w.
- [6] M. Esparza, J. Segundo, C. Gurrola-Corral, N. Visairo-Cruz, E. Barcenas, and E. Barocio, “Parameter Estimation of Grid Connected VSC Using The Extended Harmonic Domain,” *IEEE Transactions on Industrial Electronics*, no. , p. 1, 2018, doi: 10.1109/TIE.2018.2870404.
- [7] M. Esparza, J. Segundo-Ramirez, J. B. Kwon, X. Wang, and F. Blaabjerg, “Modeling of VSC-Based Power Systems in The Extended Harmonic Domain,” *IEEE Transactions on Power Electronics*, no. 99, p. 1, 2017, doi: 10.1109/TPEL.2017.2658340.
- [8] M. Esparza, J. Segundo-Ramirez, C. Nunez, X. Wang, and F. Blaabjerg, “A Comprehensive Design Approach of Power Electronic-Based Distributed Generation Units Focused on Power Quality Improvement,” *IEEE Transactions on Power Delivery*, no. 99, p. 1, 2016, doi: 10.1109/TPWRD.2016.2584616.
- [9] M. Esparza and J. Segundo-Ramírez, “An exact method for analysis and component design of grid connected VSC-based power devices,” *International Journal of Electrical Power and Energy Systems*, vol. 78, pp. 489–498, June 2016, doi: 10.1016/j.ijepes.2015.11.108.
- [10] M. Esparza and J. Segundo-Ramírez, “Optimal design of power electronics converters using the extended harmonic domain,” in *2014 IEEE International Autumn Meeting on Power, Electronics and Computing (ROPEC)*, Nov. 2014, pp. 1–6. doi: 10.1109/ROPEC.2014.7036301.

- [11] M. Esparza and J. Segundo-Ramírez, "Switching based modeling of grid interconnected hysteresis current controlled VSC converters," in *2016 IEEE International Autumn Meeting on Power, Electronics and Computing (ROPEC)*, Nov. 2016, pp. 1–6. doi: 10.1109/ROPEC.2016.7830589.
- [12] J. Kwon *et al.*, "Measurement of phase dependent impedance for 3-phase diode rectifier," in *IECON 2016 - 42nd Annual Conference of the IEEE Industrial Electronics Society*, Oct. 2016, pp. 7034–7039. doi: 10.1109/IECON.2016.7793253.
- [13] M. A. Esparza *et al.*, "Real-time emulator of an induction motor: FPGA-based implementation," in *2012 9th International Conference on Electrical Engineering, Computing Science and Automatic Control (CCE)*, Sept. 2012, pp. 1–6. doi: 10.1109/ICEEE.2012.6421152.